Diagram Pre-Staging Infrastructure for Keeper Authorship (v0.1)

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Diagram Pre-Staging Infrastructure for Keeper Authorship

Purpose

Per Casey's pedagogical directive (Saturday 14:00 EDT): the 16-volume curriculum is a "living document" with diagrams as core component. Per Keeper Team Queue 14:22 EDT (Elie NEW primary rail): pre-stage diagram generation infrastructure so Keeper can request specific diagrams during authorship and Elie turns them around quickly.

This document: 1. Inventories existing toy-generated plots that could illustrate chapters 2. Identifies high-value diagram opportunities per chapter 3. Establishes generation pipeline for new diagrams on Keeper request 4. Catalogs available tools for visualization

High-value diagram opportunities by volume

Vol 0 — Substrate Foundation

Diagram	Purpose	Tooling	Status
D_IV^5 symmetric domain geometry	Show the 10D bounded domain + isotropy K=SO(5)×SO(2)	matplotlib 3D projection	Ready to generate
Bergman kernel singularity	Visualize substrate Bergman propagator	matplotlib contour	Ready

Diagram	Purpose	Tooling	Status
5-integer cascade tree	rank=2 → N_c=3 → n_C=5 → C_2=6 → g=7 → N_max=137	graphviz	Ready
K-type Casimir spectrum	Eigenvalue ladder for K-type representations	matplotlib bar chart	Ready
4-Zone Commitment Cycle	Substrate operation cycle visualization	graphviz	Ready

Vol 1 — QFT from D_IV^5

Diagram	Purpose	Tooling	Status
Yang-Mills connection + curvature	Bundle-theoretic illustration	matplotlib	Ready
Operator zoo K-type lattice	14 operators in substrate K-rep grid	matplotlib	Ready
Renormalization flow	$\alpha(\mu)$ running with energy	matplotlib	Ready
5-family Bridge Object architecture	Heegner-trio + χ=24 + N_max + K3 + Q^5	graphviz	Ready

Vol 2 — Particle Physics

Diagram	Purpose	Tooling	Status
Standard Model particle table BST-annotated	All 27 particles with substrate origin	matplotlib table	Ready
Lepton mass ratios BST-decomposed	m_μ/m_e, m_τ/m_e cascade	matplotlib	Ready
α universal α-analog across 25 HSDs	T2462 uniqueness diagram	matplotlib	Ready
Five Absences set	5 substrate-forbidden phenomena	graphviz	Ready

Diagram	Purpose	Tooling	Status
CKM Jarlskog substrate origin	Substrate triangle visualization	matplotlib	Ready
Higgs SSB $SU(2)_L \times U(1)_Y \to U(1)_{em}$	EW symmetry breaking	graphviz	Ready

Vol 3 — Nuclear/Atomic Physics

Diagram	Purpose	Tooling	Status
SEMF coefficients table	a_V, a_S, a_C, a_A, a_P with BST primary	matplotlib table	Ready
Magic numbers visualization	2, 8, 20, 28, 50, 82, 126 substrate sources	matplotlib	Ready
Atomic orbital sequence (2l+1) = 1, N_c, n_C, g	s, p, d, f orbital shapes	matplotlib 3D	Ready
Lamb shift $\alpha^{n_C} = \alpha^5$ ceiling	5-loop QED scaling	matplotlib	Ready
Cs-137 hyperfine splitting	F=3 to F=4 transition	matplotlib	Ready

Vol 4 — GR/Cosmology

Diagram	Purpose	Tooling	Status
Λ-Casimir vacuum unification (T2418)	Substrate vacuum at multi-scale	matplotlib	Pending Lyra
Hubble four routes	Cross-validation of H_0	matplotlib	Pending Lyra
CMB cascade fingerprint n_s = 1-5/137	Spectral index BST connection	matplotlib	Pending Lyra
Cosmological cycle (Interstasis)	Big Bang cycle hypothesis	graphviz	Pending

Vol 5 — QM

Diagram	Purpose	Tooling	Status
Born rule from substrate (T2401)	Probability emergence	matplotlib	Pending Lyra
Decoherence master equation (T2480)	Substrate mechanism	matplotlib	Pending Lyra
Bell inequality + Tsirelson	BST sub-Tsirelson 1/8	matplotlib	Ready

Vol 7 — Electromagnetism

Diagram	Purpose	Tooling	Status
Maxwell equations + Yang-Mills derivation	Substrate Bergman bundle	graphviz	Ready
α^{BST primary} exponent ladder (T2476)	Lamb + a_e + hyperfine	matplotlib	Ready
EM field tensor Lorentz transformation	E ↔ B mixing	matplotlib	Ready
Multipole expansion (2l+1)	Substrate orbital parallel	matplotlib	Ready
Photonic crystal $10K experiment	Lattice + band gaps	matplotlib	Ready

Vol 8 — Classical Mechanics

Diagram	Purpose	Tooling	Status
Phase space + Hamilton flow	Symplectic structure	matplotlib	Ready
Lorenz attractor (chaos)	Standard chaos visualization	matplotlib	Ready
Kepler orbits + Runge-Lenz vector	Closed orbit hidden symmetry	matplotlib	Ready
Inertia tensor + Euler angles	Rigid body dynamics	matplotlib 3D	Ready

Vol 9 — Condensed Matter

Diagram	Purpose	Tooling	Status
BaTiO3 137-plane crystallography	$25K BST falsifier visualization	matplotlib 3D	Ready
Photonic crystal band structure	$10K falsifier visualization	matplotlib	Ready
Kitaev honeycomb model	Spin liquid + anyons	matplotlib	Ready
B12H32 superconductor T_c~214K	Hydride structure	matplotlib	Ready

Vol 10-13 — Math + Special Domains

Diagram	Purpose	Tooling	Status
Heegner number landscape	-1, -2, -3, -7, …, -163 with BST primary	matplotlib	Ready
K3 surface Hodge diamond	Cohomology table	matplotlib	Ready
Monster group character table	Moonshine relations	matplotlib table	Ready
Periodic table BST-annotated	Vol 12 chemistry	matplotlib table	Ready
Genetic code substrate-derivation	Vol 13 biology	matplotlib table	Ready

Vol 14-15 — Information Theory + Methodology

Diagram	Purpose	Tooling	Status
Shannon channel capacity BST	Vol 14 info theory	matplotlib	Ready
AC theorem graph topology	Vol 15 methodology	graphviz	Ready
Methodology stack (20 layers)	Vol 15 audit framework	graphviz	Ready
Calibration #1-#24 timeline	Vol 15 governance	matplotlib	Ready

Inventory of existing toy-generated plots

Toy #	Plot type	Chapter relevance
Toy 273-278	Seeley-DeWitt heat kernel	Vol 1 Ch 5 (Casimir) + Paper #9
Toy 305 + 361	Speaking pair period	Vol 1 Ch 5 + Paper #9
Toy 612-639	k=16 ratio = -dim SU(5)	Vol 2 Ch 5 (Gauge hierarchy)
Toy 541	Five integers to everything	Vol 0 Ch 4 + reproduction package
Toy 1401	n_s = 1-5/137 CMB	Vol 4 Ch 4 (Cosmology)
Toy 1410	Discrete Gauss-Bonnet	Vol 0 Ch 7 + P≠NP
Toy 1543	Null-model 3σ above random	Vol 0 Ch 9 + Strong-Uniqueness
Toy 3221 + 3222	5-family Bridge Object	Vol 0 Ch 7 + Master Doc
Toy 3504	Conservation laws	Vol 8 Ch 5
Toy 3505	Yang-Mills connection	Vol 7 Ch 2 + Vol 1 Ch 8
Toy 3506	Decoherence + per-BC	Vol 5 Ch 10
Toy 3507	Bogoliubov-GF(128)	K52a Session 7 paper-grade
Toy 3508	SP-30-3 algebraic verification	SP-30-3 v0.1 proposal

Generation pipeline

Python tools available: - matplotlib 3.x with mathtext + LaTeX support - mpmath for high-precision numerical work - networkx for graph visualizations - graphviz (system binary + python bindings) - numpy + scipy for computation

LaTeX integration: - xelatex + STIX Two Text fonts in PDFs (Curriculum + paper-grade notes) - mathjax/MathML for any web-rendered diagrams

Typical generation time: - Simple matplotlib plot: 5-15 min including iteration - Complex 3D visualization: 30-60 min - Network/graph diagram: 15-30 min - Composite figure (subplot multi-panel): 30-90 min

Output formats: - PNG (300 DPI) for embedding in markdown - PDF for high-quality printing - SVG for web/interactive applications

Workflow when Keeper requests a diagram

1. Keeper signals in broadcast: "Vol 0 Ch 1 needs D_IV^5 symmetric domain illustration"
2. Elie checks inventory: existing toy plot adaptable, or new generation needed
3. Elie generates: 5-90 min depending on complexity

4. Output: figure file + LaTeX/markdown embedding code + caption suggestion
5. Keeper integrates into authorship + cites toy/data source

Initial priorities

Per Keeper's authorship cadence (Vol 0 first, then 1, 2, …): pre-stage Vol 0-2 diagrams. These are LIKELIEST to be requested first:

Vol 0 priorities: - D_IV^5 symmetric domain visualization (likely Vol 0 Ch 1) - 5-integer cascade tree (likely Vol 0 Ch 4) - Bergman kernel singularity (likely Vol 0 Ch 5) - 4-Zone Commitment Cycle (likely Vol 0 Ch 8 or 11)

Vol 1 priorities (when Vol 0 author-pass complete): - Operator zoo K-type lattice - Yang-Mills connection + curvature - 5-family Bridge Object architecture

Vol 2 priorities: - Standard Model particle table BST-annotated - α universal α-analog 25 HSDs comparison - Five Absences set

Cal #50 + Cal #99 compliance

INTERNAL document. External-facing diagram captions use operational language only: - "BST identifies / BST predicts / BST derives" - Cal #99 META-theorem framing preserved - No substrate-cognition framing externally

Bibliography

1. Casey pedagogical directive (Saturday 14:00 EDT): living document + diagrams.
2. Keeper Team Queue Pre-Authorship (Saturday 14:22 EDT): Elie NEW primary rail.
3. Inventory derived from `play/` toy outputs + existing Curriculum/ chapters.

— Elie, Diagram Pre-Staging Infrastructure v0.1, 2026-05-23 Saturday 15:00 EDT (`date`-verified)